INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

BUSINESS CONTINUITY AND DISASTER RECOVERY

GRC205 — Incident Management and Business Continuity | Week 21

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Scenario: Northbridge Skills Institute (fictional training institute)

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TASK 1: BUSINESS IMPACT ANALYSIS (BIA)

1.1 Introduction

A Business Impact Analysis (BIA) helps an organisation understand how a disruption affects its important activities and what needs to be prioritised to keep operations going. It also helps management identify the resources needed to maintain essential services while normal operations are being restored.

For Northbridge Skills Institute, flooding and power failure make the main site and its local systems unavailable from 08:00 WAT on Monday, Day 0. The site is expected to remain inaccessible for 72 hours. The disruption affects teaching and assessment, fees and finance, admissions, staff payroll, and records and certificates.

In carrying out this assessment, I have considered the financial impact alongside the effect on learners and staff, operational deadlines, data integrity and the institute's reputation. The proposed priorities are based on the information provided in the scenario and may need to be reviewed as the situation develops.

1.2 Business Impact Analysis of the Five Functions

The table below summarises the five functions, their owners, the people and resources involved, their dependencies, the expected impact of disruption and the minimum service that should be maintained.

Function and proposed owner	Users and impact over time	Critical resources, premises, suppliers and information	Dependencies	Minimum acceptable service and proposed business priority
F1: Teaching and assessment — Academic/Examinations Coordinator	120 learners scheduled for the 11:00 examination, examination staff and academic management. The immediate concern is the examination	LMS01, DB01, examination personnel, secure assessment records, learner identity information, the alternate facility and suitable devices. The main	Recovery network, AUTH01, DB01, LMS01, available technical and examination staff, secure access and communication.	**Priority 1 — Immediate.** Arrange an approved examination solution for all 120 learners. The 25 workstations can accommodate part

	deadline. If no acceptable assessment service is available by 11:00, a one-off rescheduling cost of NGN 120,000 applies. Continued disruption affects learner experience and assessment schedules.	site is inaccessible; 25 alternate-site workstations are available from 10:00.		of the centre-based requirement, while an approved solution must be found for the remaining 15 learners who need centre workstations. The 80 learners who can use their own devices may do so if the assessment service and security arrangements support it. Record attendance and outcomes securely.
F2: Fees and finance — Finance Manager	Learners, finance staff and management. Fee confirmation is due by 16:00 on Day 0. As disruption continues, payment records may remain unupdated and reconciliation work may increase.	FIN01, DB01, finance staff, transaction records, local receipts and wallet records, and the external payment provider. The provider remains available, but local records cannot be updated during the outage.	Recovery network, AUTH01, DB01, FIN01, authorised staff, secure access and communication.	**Priority 2 — High.** Maintain a controlled record of payments and transactions awaiting entry. Use verified provider evidence and reconcile temporary records against restored records before closing outstanding items.
F3: Admissions and registration — Admissions/Registry Coordinator	Applicants, prospective learners, registry staff and management. Admissions close at 18:00 on Day 0. Continued disruption prevents new applications and registration changes from being entered into the local system.	REG01, DB01, admissions staff, application and registration information, and communication channels. The main site is unavailable, so a temporary submission arrangement may be necessary.	Recovery network, AUTH01, DB01, REG01, staff, secure access and communication.	**Priority 3 — High.** Provide an approved temporary way to receive applications and registration changes, maintain a controlled log and ensure that records are entered once the system is available.
F4: Staff payroll — HR/Payroll Manager	Staff, HR and payroll personnel,	HR01, DB01, verified payroll	Recovery network,	**Priority 4 — High, with**

	finance management and the receiving bank or payroll service provider. The submission deadline is 12:00 on Day 1. Incorrect or duplicate instructions could affect staff and require manual correction.	information, authorised payroll staff, bank submission information and secure communication.	AUTH01, DB01, HR01, available staff, secure access and communication.	**preparation beginning early.** Protect payroll records and prepare the submission using verified information. Any temporary processing must be authorised, checked and reconciled. No unverified or duplicate payment instructions should be submitted.
F5: Records and certificates — Records/Registry Manager	Learners, graduates, records staff and other users requesting records or certificates. The internal service commitment is 48 hours. Longer disruption may delay requests and affect confidence in the institute's record services.	FS01, certificate and learner files, records staff, secure storage and access controls. The main site is unavailable, so authorised staff may need a temporary process for urgent requests.	Recovery network, AUTH01, FS01, available staff, secure access and communication.	**Priority 5 — Planned recovery, with urgent requests assessed individually.** Maintain a log of outstanding requests and handle urgent cases through an approved process where possible. Protect file integrity and access restrictions throughout the disruption.

The priority sequence is a proposed business ranking, not a statement that the systems must be restored in that same order. It is based on the deadlines and consequences in the scenario. Payroll preparation should begin before its Day 1 deadline even though F4 is ranked below the services with Day 0 deadlines. Similarly, urgent records requests should still be assessed rather than automatically postponed.

The proposed temporary services also require approval from the relevant business owner. For any manual process, the institute should keep a dated record, restrict access to authorised personnel, obtain the required approvals and reconcile the record against the restored system.

1.3 Estimated Financial Impact of Disruption

The assignment provides an hourly disruption cost for each function. For this exercise, the costs are treated as separate, non-overlapping incremental impacts. The one-off examination rescheduling cost of NGN 120,000 is added when the 11:00 trigger is met.

The calculation is:

Hourly disruption cost × number of hours unavailable

For F1, the NGN 120,000 rescheduling cost is added separately. The calculations assume no recovery or continuity measures and that the five functions incur their stated hourly impacts throughout the relevant period.

A. Estimated impact after 8 hours

Function	Calculation	Estimated impact
F1: Teaching and assessment	(NGN 60,000 × 8) + NGN 120,000	NGN 600,000
F2: Fees and finance	NGN 40,000 × 8	NGN 320,000
F3: Admissions and registration	NGN 20,000 × 8	NGN 160,000
F4: Staff payroll	NGN 10,000 × 8	NGN 80,000
F5: Records and certificates	NGN 5,000 × 8	NGN 40,000
Total estimated impact		**NGN 1,200,000**

The estimated impact after eight hours is NGN 1.2 million, including the examination rescheduling cost. This assumes that no acceptable examination service is available by 11:00 and that the rescheduling trigger is therefore met.

B. Estimated impact after 24 hours

Function	Calculation	Estimated impact
F1: Teaching and assessment	(NGN 60,000 × 24) + NGN 120,000	NGN 1,560,000
F2: Fees and finance	NGN 40,000 × 24	NGN 960,000
F3: Admissions and registration	NGN 20,000 × 24	NGN 480,000
F4: Staff payroll	NGN 10,000 × 24	NGN 240,000
F5: Records and certificates	NGN 5,000 × 24	NGN 120,000
Total estimated impact		**NGN 3,360,000**

The estimated 24-hour impact is NGN 3.36 million. F1 contributes NGN 1.56 million, including the one-off rescheduling cost, while F2 contributes NGN 960,000. These figures help show how the impact can accumulate when services remain unavailable.

C. Limitations of the estimates

The first limitation is that the hourly amounts are exercise estimates, not observed losses or confirmed cash payments. The actual financial impact would depend on the duration of disruption, the effectiveness of temporary arrangements and the additional work required to restore normal operations.

The second limitation is that the calculations do not capture every possible consequence. For example, payroll disruption could lead to incorrect or duplicate payments, reconciliation work and additional costs to correct errors. These consequences could increase the overall financial impact. The figures also do not

fully reflect loss of learner confidence, damage to the institute's reputation or the welfare of learners and staff.

Data integrity is another concern. A system may be available after restoration, but its records must still be checked for completeness and accuracy before the institute relies on them. The NGN 3.36 million estimate should therefore be treated as an indicator of the potential disruption impact, not a guarantee of the total cash loss.

1.4 Business Priority Versus Technical Restoration Order

Business priority is not determined by cost alone. The institute must also consider the time of each deadline, the number of people affected, the consequences of delay, the reliability of information and the work required to reconcile transactions.

F1 is the first business priority because the examination starts at 11:00 on Day 0. The institute should immediately arrange an acceptable examination service instead of waiting for the normal system to be restored. The alternate facility has 25 workstations, while 40 learners need centre workstations. This leaves 15 learners requiring another approved arrangement. The other 80 learners may use their own devices if the assessment service and security arrangements permit it.

F2 is the next priority because fee confirmation is due at 16:00 on Day 0. Although the external payment provider remains available, the local receipts and wallet records cannot be updated. The institute must maintain reliable temporary records and plan for reconciliation. F3 follows because admissions close at 18:00 on Day 0. F4 has a later deadline, but payroll preparation should start early because incorrect or duplicate instructions could affect staff. F5 has a 48-hour internal service commitment, and urgent requests should be considered throughout the disruption.

However, the technical restoration sequence is governed by dependencies. F1 to F4 require the recovery network, AUTH01, DB01 and their respective applications. F5 requires the recovery network, AUTH01 and FS01. The team must restore and validate shared components before proceeding with the applications that depend on them.

This means that the business team and technical team need to coordinate. Management and business coordinators should decide which services are most urgent and arrange temporary alternatives, while the technical team follows the necessary restoration sequence and validates systems before they are used.

1.5 Important Dependency Risks

1. Failure of shared infrastructure

F1 to F4 depend on the recovery network, AUTH01 and DB01. If any of these components fails to restore successfully, several services could remain unavailable at the same time. This could increase disruption and delay services with urgent deadlines. The technical team should validate shared components and have an escalation process for failed recovery steps.

2. Limited technical staff and competing recovery activities

Only two disaster recovery technicians are available. Recovery tasks cannot be split, and parallel work is possible only when prerequisites are complete and technicians are available. If several systems require

attention at the same time, limited staffing could delay restoration. The recovery schedule must therefore reflect technical dependencies while supporting the agreed business priorities.

3. Secure access and communication dependencies

All five functions depend on available staff, secure access and communication. If staff cannot access the restored systems securely, or if learners and other stakeholders do not receive clear instructions, the institute could experience further delays, confusion or unauthorised access. The institute should use verified communication channels and maintain appropriate access controls over systems and temporary records.

1.6 Conclusion

The BIA shows that Northbridge Skills Institute must address its most time-sensitive services while preparing for a recovery that depends on shared infrastructure. The examination requires immediate attention because of the 11:00 deadline, but finance, admissions, payroll and records also require controlled temporary arrangements.

The estimated financial impact increases from NGN 1.2 million after eight hours to NGN 3.36 million after 24 hours under the stated assumptions. These estimates support the need for timely action, but they do not capture every financial, operational and reputational consequence.

My proposed approach is to manage the business priorities according to deadlines and wider impact while allowing the technical team to restore systems in the correct dependency order. This should help the institute protect learners and staff, maintain reliable records and resume normal services in a controlled manner.

Task 2 — Recovery Objectives and Backup Design

2.1 Recovery Objectives and Justification

Following the disruption at Northbridge Skills Institute, it is important to determine how quickly each business function should be restored and how much data the institute can afford to lose. Not all functions have the same level of urgency, so the recovery targets should be based on their business importance, deadlines and the possible consequences of a prolonged outage.

The Recovery Time Objective (RTO) is the target time within which a business service should be restored after a disruption. The Recovery Point Objective (RPO) refers to the maximum amount of recent data the organisation can afford to lose. Maximum Tolerable Downtime (MTD) is the longest period a business function can remain unavailable before the consequences become unacceptable.

One important point is that RTO and RPO are not the same. A system may take longer to restore, but the organisation may still need to protect its data against significant loss. For example, the records system may have a 24-hour recovery target, while its data-loss target remains 30 minutes because the institute needs current and accurate information to process certificate requests.

Based on the information provided in the scenario, I propose the following recovery objectives for the five business functions. These are recommended targets that should be reviewed and tested before they are adopted as official commitments.

Function	Proposed RTO	Proposed RPO	Proposed MTD	Reason for the target
F1 – Teaching and assessment	2 hours	30 minutes	3 hours	The examination starts at 11:00, so recovery is urgent. The short RTO is challenging and will require a tested temporary assessment arrangement if normal recovery takes longer.
F2 – Fees and finance	4 hours	30 minutes	8 hours	Restoring the system by noon would leave four hours before the 16:00 fee-confirmation deadline for payment reconciliation and backlog processing.
F3 – Admissions and registration	5 hours	30 minutes	10 hours	Restoring the system by 13:00 would leave five hours before admissions close at 18:00 to enter and validate outstanding applications.
F4 – Staff payroll	12 hours	30 minutes	28 hours	This is a proposed target that allows time to restore and check payroll records before the bank submission deadline at 12:00 on Day 1.
F5 – Records and certificates	24 hours	30 minutes	48 hours	Restoring the records service within 24 hours would leave additional time before the 48-hour internal service commitment. Current and accurate records are still necessary for certificate processing.

The targets are based on the business requirements in the scenario, but some assumptions require further confirmation. In particular, the payroll owner should confirm whether the proposed F4 targets are realistic. The MTD values for F2 and F3 should also be confirmed with the relevant business owners because their deadlines do not automatically establish the full period of downtime they can tolerate.

For F1, the two-hour RTO is especially challenging. The warm recovery option estimates that the first critical service may take between three and five hours to restore. Northbridge therefore cannot depend on the warm recovery arrangement alone to meet the two-hour target. A temporary examination arrangement would need to be prepared and tested, or the target would need to be formally reviewed.

When should a service be declared recovered?

In my view, a service should not be declared recovered simply because the server has started or the application can be opened. The institute must also confirm that the service is working properly and can be used safely.

Before declaring a function recovered, the recovery team should confirm that the required network and dependencies are available, the application or file service works as expected, and the restored data is complete and consistent. Access permissions and other security controls should also be checked. Where temporary records have been used, the team must ensure that outstanding transactions, applications or requests are properly tracked and reconciled.

The relevant business owner should confirm that the restored service is acceptable for use. The time required for these checks and for processing outstanding work must be included in the recovery plan.

The difference between the RTO and MTD provides some room for validation and unexpected problems. However, this buffer does not guarantee that every outstanding request will be completed within the remaining time.

2.2 Checkpoint Calculations and Potential Data Loss

The disruption occurred at 08:00 WAT on Day 0. To determine the potential data-loss window, I calculated how old each latest stated backup or recovery checkpoint was at the time of the disruption.

System	Latest stated checkpoint	Calculation	Age at 08:00	Functions affected
DB01	Incremental chain ending at 07:00 on Day 0	08:00 − 07:00	1 hour	F1, F2, F3 and F4
AUTH01	Snapshot at 00:00 on Day 0	08:00 − 00:00	8 hours	F1–F5
FS01	Full backup at 23:00 on Day −1	08:00 − 23:00 on Day −1	9 hours	F5

These figures show the age of the latest stated checkpoints. They indicate the potential gap between the disruption and the available recovery point, but they do not automatically mean that all data changed during those periods has been lost.

DB01

The latest stated DB01 recovery point is the incremental chain ending at 07:00 on Day 0. To restore the database to this point, the team needs the B01 full backup taken at 23:00 on Day −1 and B02, the incremental backup at 07:00 on Day 0. It must also confirm that every required intermediate incremental backup is available and valid.

The exact restoration sequence must be confirmed from the backup catalogue and the database recovery documentation. If a required component is missing or corrupt, the team may have to recover to an earlier point.

If the chain is complete and passes integrity checks, the latest stated DB01 checkpoint is one hour before the disruption. However, the proposed RPO is 30 minutes. This means the stated checkpoint is 30 minutes older than the proposed target and does not, on its own, demonstrate that the RPO can be met.

DB01 supports teaching and assessment, fees and finance, admissions and registration, and staff payroll. A problem with its recovery can therefore affect several business functions at once.

AUTH01

The latest stated AUTH01 snapshot was taken at midnight on Day 0, eight hours before the disruption. Since all five functions depend on AUTH01, the institute must consider its recovery alongside the database and application recovery.

Even if an application or database has been restored, staff may still be unable to access it if the local directory service is unavailable or its configuration is incorrect. The team must therefore validate authentication, user access and security controls before allowing the restored services to return to normal use.

The eight-hour-old snapshot also means that the current recovery arrangement does not demonstrate compliance with the proposed 30-minute RPO.

FS01

The latest stated FS01 backup was taken at 23:00 on Day −1, which is nine hours before the disruption. If the institute has to recover from this backup alone, changes made after that time may not be present in the restored files.

This is important because FS01 supports records and certificates. Even though the internal service commitment is 48 hours, the institute still needs accurate and reasonably current records to process certificate requests correctly. For this reason, I propose an RPO of 30 minutes, although the existing checkpoint does not demonstrate that this target can be achieved.

Effect on the business functions

The checkpoint analysis identifies different risks for the affected functions.

- **F1 – Teaching and assessment:** Recent assessment records may need to be checked after recovery. Any temporary examination arrangement must protect fairness, confidentiality and assessment integrity.
- **F2 – Fees and finance:** Successful payments recorded by the external payment provider may help the institute reconcile transactions missing from the restored finance database. The team must verify the records and avoid duplicate entries.
- **F3 – Admissions and registration:** Temporary application records or approved intake logs may help staff re-enter missing applications and registration changes. However, their existence and completeness cannot be assumed.
- **F4 – Staff payroll:** Payroll inputs and approved payroll records may help the team validate the restored data before submitting instructions to the bank. Missing information must not be reconstructed without reliable evidence.
- **F5 – Records and certificates:** Other authorised records may help verify files that are missing or outdated after restoration, but their availability and accuracy must be confirmed.

The disconnected full backup taken at 17:00 on Day −3 provides another possible recovery point. However, it is older than the other stated checkpoints and would not meet the proposed RPOs. It may still be useful if more recent copies are unusable, provided the backup is checked and approved for recovery.

Temporary records can help the institute reconcile missing information, but they should not automatically be treated as complete replacements for lost system data. The institute must verify their accuracy, timestamps and completeness before relying on them.

2.3 Proposed Backup Schedule and Restoration Design

The checkpoint analysis shows that the current backup arrangement does not demonstrate that the proposed 30-minute RPOs can be achieved. I would therefore recommend a backup schedule that provides more frequent recovery points and maintains enough backup history to support recovery if a recent copy is damaged or unusable.

The following is a proposed schedule. The actual implementation will depend on the capabilities of the database, directory and backup systems.

System or backup type	Proposed schedule	Proposed retention	Purpose
DB01 full backup	Nightly	30 daily copies	Provides a baseline for database restoration.
DB01 transaction-log backups, where supported	Every 15 minutes	At least 7 days of logs, subject to recovery and retention requirements	Supports point-in-time recovery and provides a margin within the 30-minute RPO.
DB01 additional full backup	Weekly	8 weekly copies	Provides another recovery point if a recent daily backup chain is unusable.
AUTH01 protected recovery copy	At intervals no longer than 30 minutes, using a suitable and tested method	30 daily recovery points and 8 weekly copies	Reduces the current recovery gap for identity services.
FS01 full backup	Nightly	30 daily copies	Provides a complete baseline for records and certificate files.
FS01 incremental backup	Every 15 minutes, subject to validation	7 days of incremental copies, alongside the required full backups	Supports the proposed 30-minute RPO for records.
Independent protected copy	At least daily, with additional protection for critical recent recovery points	Daily and weekly copies according to the approved policy	Provides another recovery option if production or the main backup environment is compromised.

The retention periods above are recommendations rather than requirements stated in the scenario. Northbridge should confirm the final periods based on storage capacity, records-retention obligations, privacy requirements, recovery needs and cost.

The proposed 15-minute backup intervals are intended to provide some margin within the 30-minute RPO. However, frequent backups alone do not guarantee that the target can be achieved. The backup jobs must complete successfully, the required recovery components must be available, and restoration must be tested.

For DB01, transaction-log backups can only be used as proposed if the database is configured to support the required recovery model. The database administrator must

confirm this. If the configuration does not support it, Northbridge will need another tested method that can meet the approved RPO.

Applying the 3-2-1 backup principle

Northbridge should apply the 3-2-1 backup principle to reduce the risk of losing all available copies during a major incident.

- **Three copies:** Keep the production data and at least two additional backup copies.
- **Two storage types:** Store the copies using two distinct storage types or platforms, reducing dependence on one environment.
- **One offsite copy:** Keep at least one copy at a separate location so that a disaster at the main site does not destroy every copy.

In addition, I recommend maintaining an independent offline or immutable copy. This is important because an incident affecting production could also affect connected backups, especially where administrative access is shared.

Backup data should be encrypted both in transit and at rest. Encryption keys should be protected separately, and access to backups should be limited to authorised personnel. Separate backup-administration accounts, least-privilege access and multifactor authentication should be used where supported. Backup deletion and changes to retention settings should also be restricted and logged.

These controls address a concern identified in the scenario: production and network backup administration share an administrator identity, while the offsite provider's access-independence controls are undocumented. Keeping an encrypted copy offsite is not enough if the same compromised account can alter or delete it.

2.4 Backup Integrity, Access Independence and Restoration Testing

Having backups does not necessarily mean that Northbridge can recover its services successfully. An offsite backup could be incomplete, corrupted, inaccessible or unusable because its encryption key is unavailable. Similarly, a live replica may copy accidental deletion, corrupted data or malicious changes from the production system.

For this reason, Northbridge should verify the integrity of its backups, confirm that backup access is independent of production administration, and regularly test whether the restored systems and data can actually support business operations.

Control area	What Northbridge should do	Evidence to retain
Backup integrity	Check backup completion, integrity and catalogue consistency. Confirm that the required DB01 restore chain is complete.	Backup reports, integrity-check results and records of any failures.

Access independence	Confirm that production administrators cannot independently delete or alter every backup copy. Test access using separate recovery credentials.	Access-control reviews, privileged-access logs and documented test results.
Encryption	Confirm that backups are encrypted and authorised recovery staff can retrieve the necessary keys.	Encryption configuration and successful key-retrieval tests.
Restoration testing	Restore DB01, AUTH01 and FS01 in an isolated test environment using documented procedures.	Restoration times, recovery points achieved, errors and test results.
Data and service validation	Check database consistency, authentication, application functionality, file permissions and sample business transactions.	Validation records and business-owner approval.
Corrective actions	Record failures, assign owners and deadlines, and repeat failed tests after corrective work.	Corrective-action register and evidence of successful retesting.

Proposed restoration testing programme

I recommend that Northbridge carry out regular automated checks of backup jobs and integrity, where supported, and conduct a full recovery exercise at least quarterly. The testing frequency should be reviewed according to the risk and the results of previous tests.

For DB01, the recovery team should restore the full backup and apply the required incremental or transaction-log backups in the correct order. The team should record the recovery point achieved and compare it with the approved RPO.

For AUTH01, the test should confirm that users can authenticate and that access controls work correctly. For FS01, the team should confirm that sample records can be opened, permissions remain correct, and authorised staff can use the recovered files.

The actual restoration time should be compared with the approved RTO. Business owners should validate the restored service before it is declared recovered. Where a test fails, the failure should be documented, an owner assigned, and a new test conducted after the problem has been corrected.

Assumptions that require confirmation

Before implementing this design, Northbridge should confirm the following:

1. DB01 supports the proposed backup method and point-in-time recovery capability.
2. The full backup and all required incremental or transaction-log backups are available and form a valid restoration chain.

3. AUTH01 and FS01 support the proposed backup intervals and recovery methods.
4. Sufficient storage, network capacity, licensing and technical staff are available to operate the schedule.
5. The offsite provider can demonstrate independent access controls and successful restoration of encrypted copies.
6. Temporary payment, admissions and payroll records are available where needed and can be validated by the relevant business owners.

These assumptions should be checked through technical assessment, supplier evidence and restoration tests. Until then, the proposed schedule should be treated as a recovery design rather than a proven capability.

Conclusion

The proposed recovery objectives are intended to protect Northbridge's important business functions while recognising their different operational deadlines. The checkpoint calculations show that DB01's latest stated recovery point is one-hour old, AUTH01's snapshot is eight hours old, and FS01's backup is nine hours old at the time of disruption. These checkpoints do not demonstrate that the proposed 30-minute RPOs can be achieved.

To address the gaps, Northbridge should introduce more frequent backups, maintain a verified DB01 restoration chain, protect independent backup copies, restrict administrative access and regularly test its recovery arrangements. Services should only be declared recovered after the technical checks, data validation and business-owner confirmation have been completed.

The main lesson is that having backups is not enough. Northbridge must be able to demonstrate that the backups are protected, usable and capable of supporting the agreed recovery objectives.

Task 3 — Recovery Strategy and Budget (15 marks)

3.1 Comparison of Recovery Options and Annual Costs

Northbridge Skills Institute has three recovery options to consider: Cold, Warm and Hot. Each option has a different cost, recovery time and potential data-loss window. The institute needs to select an option that supports its business recovery objectives while staying within the annual budget of NGN 3,000,000.

The assignment also requires an additional NGN 400,000 annually for independent protected backups and NGN 200,000 for training and testing. These costs must be included when comparing the options.

Annual cost comparison

The total annual cost is calculated by adding the base package cost, protected backup cost and training and testing cost.

Cost item	Cold	Warm	Hot
Base package	NGN 600,000	NGN 1,500,000	NGN 2,400,000
Independent protected backups	NGN 400,000	NGN 400,000	NGN 400,000
Training and tests	NGN 200,000	NGN 200,000	NGN 200,000
Total annual cost	**NGN 1,200,000**	**NGN 2,100,000**	**NGN 3,000,000**
Annual budget limit	NGN 3,000,000	NGN 3,000,000	NGN 3,000,000
Budget remaining	NGN 1,800,000	NGN 900,000	NGN 0

The calculations show that all three options fit within the stated budget when the two additional costs are included. However, the Hot package uses the entire annual budget, while Cold and Warm leave funds available for other operational needs.

Comparison of the recovery packages

Factor	Cold	Warm	Hot
Estimated first critical service	24–48 hours	3–5 hours	About 1 hour
Estimated data-loss window	Up to 24 hours	1–6 hours	Up to 15 minutes
Recovery performance	Slowest recovery	Moderate recovery time	Fastest estimated recovery
Fit with proposed RTOs	Poor for urgent functions	Better, but may miss urgent targets	Closest fit for urgent recovery targets
Fit with proposed 30-minute RPOs	Does not meet the stated target	Does not meet the stated target	The stated window is within the target
Staffing requirements	More setup and coordination may be needed during recovery	Requires coordination of the recovery activities and validation	Still requires skilled staff to validate and recover services

Geography and connectivity	Recovery location and access arrangements must be confirmed	Alternate-site location, connectivity and access must be confirmed	Location, connectivity and geographic separation must be confirmed
Supplier dependency	May depend on supplier provisioning and setup during an incident	May depend on supplier availability and recovery support	May depend on supplier availability, standby infrastructure and contractual commitments
Risk of corruption or malicious changes	Older copies may be available, but recovery may be slow	Recovery points may be available, but corruption remains a risk	A current replica may reproduce corruption quickly without independent protected copies

The recovery times and data-loss windows above are estimates supplied in the assignment. They should be treated as planning figures rather than guaranteed results. Northbridge must confirm the actual capabilities of each package, including supplier dependencies, recovery location, connectivity and staffing arrangements.

The Cold package is the least expensive, but its estimated recovery time of 24–48 hours makes it unsuitable as the main recovery solution for time-sensitive activities such as the 11:00 examination. Its potential data-loss window of up to 24 hours is also much greater than the proposed 30-minute RPOs.

The Warm package offers a middle ground between cost and recovery speed. Its annual cost of NGN 2,100,000 leaves NGN 900,000 within the budget. However, its estimated first-service recovery time of 3–5 hours would not meet the proposed two-hour RTO for teaching and assessment. Its estimated data-loss window of 1–6 hours also exceeds the proposed 30-minute RPOs.

The Hot package offers the shortest estimated recovery time and the smallest stated data-loss window. Its total annual cost is NGN 3,000,000, exactly the amount available in the budget. It therefore provides the closest fit with the institute's proposed recovery objectives, although the supplier's claims must be verified through testing.

3.2 Recommended Recovery Option

I recommend the Hot recovery package, including independent protected backups and training and testing, subject to confirmation that the supplier can demonstrate the stated recovery capability and that the institute can sustain the full annual cost.

My main reason for recommending Hot is the urgency of the institute's business activities. The examination starts at 11:00, fee confirmation is due at 16:00, admissions close at

18:00 on Day 0, and payroll instructions must be submitted by 12:00 on Day 1. Delayed recovery could affect learners, staff and the institute's ability to meet its commitments.

The Hot package has an estimated first critical service recovery time of about one hour and a potential data-loss window of up to 15 minutes. These figures align more closely with the proposed two-hour RTO for F1 and the 30-minute RPOs than the other options.

However, restoring the first critical service within one hour does not mean that all five business functions will be fully recovered within that time. Northbridge must confirm the recovery time for each application and its dependencies, including AUTH01, DB01 and the relevant application or file service. The time needed for validation and backlog processing must also be considered.

The additional investment in independent protected backups is important because the scenario identifies a shared administrator identity for production and network backup administration. If that identity is compromised, connected backups could also be at risk. Independent protected copies, together with restricted access, can reduce this exposure.

Training and testing are equally important. Staff need to understand their responsibilities and practise the recovery procedures before an actual disruption occurs. Testing can also identify problems with backup restoration, access controls, supplier arrangements and the time required to restore individual services.

Objectives the Hot package may not meet

Although Hot is the closest fit based on the available estimates, I would not assume that it meets every recovery objective without evidence.

First, the estimated one-hour recovery time refers to the first critical service, not necessarily all services. F1 has a proposed two-hour RTO, so the complete examination service must be restored and validated within that period to meet the target. Since the Warm option's estimated recovery window is longer than this, Northbridge should also prepare a tested temporary examination arrangement in case normal recovery is delayed.

Second, the Hot package's stated data-loss window of up to 15 minutes is within the proposed 30-minute RPO. However, this must be confirmed for the systems supporting the five functions. The institute should not assume that the quoted figure applies equally to every application, database, directory and file service.

Finally, the Hot package leaves no remaining budget allowance. Any additional costs for connectivity, licensing, storage, supplier support or implementation must be identified before approval.

Residual risks and responsibility for acceptance

Even with Hot, some risks will remain. These risks should be documented, managed and accepted by the appropriate authority.

Residual risk	Proposed treatment	Responsibility for accepting the remaining risk
Supplier may not achieve the quoted recovery times	Request evidence from recovery tests, clear contractual service levels and defined escalation arrangements	Senior management, with advice from the recovery coordinator and technical lead
A recent backup or replica may contain corrupted or malicious data	Maintain independent offline or immutable copies and validate recovery points before use	Senior management, advised by technical and security leads
Only two technicians are available for recovery	Document procedures, prioritise activities and train additional personnel where possible	Technical lead and senior management
The Hot package uses the full annual budget	Confirm all recurring and additional costs before procurement	Budget owner and authorised approving officer
External connectivity or supplier services may be unavailable	Confirm supplier dependencies and maintain alternative communication and connectivity arrangements	Senior management and the relevant service owner

Risk acceptance should be recorded formally. The approving authority should understand which recovery targets are supported by evidence, which remain uncertain and what the consequences could be if recovery takes longer than planned.

3.3 Comparison with Estimated Disruption Costs

The financial estimates from Task 1 provide a useful basis for considering the value of recovery investment.

Scenario	Estimated disruption cost
Eight-hour disruption, including the one-off examination rescheduling cost	NGN 1,200,000
Twenty-four-hour disruption, including the one-off examination rescheduling cost	NGN 3,360,000

Hot package annual cost, including protected backups and training and tests	NGN 3,000,000

The estimated cost of an eight-hour disruption is NGN 1.2 million, while the estimated cost of a 24-hour disruption is NGN 3.36 million. The latter is NGN 360,000 higher than the Hot package's total annual cost.

This comparison shows why recovery capability is important, but it does not prove that purchasing Hot will save NGN 360,000 or guarantee a positive return on investment. The disruption figures are estimates based on the scenario, not actual losses experienced by the institute. They may also exclude other consequences, such as reputational damage, additional staff time, learner complaints and the effects of incorrect payroll or finance records.

A single-outage estimate cannot establish annual return on investment because it does not show how often a disruption is likely to occur, how long an outage would last under each package or how effectively the package would reduce the impact. The institute also needs to consider the full cost of implementation, ongoing support and any additional infrastructure or supplier charges.

Evidence required before approval

Before approving the Hot package, I would request the following evidence:

1. **Supplier recovery-test results:** Evidence showing actual recovery times and the data-loss windows achieved for the systems in scope.
2. **Contractual commitments:** Clear service levels, escalation arrangements, recovery support, exclusions and any additional charges.
3. **Technical architecture:** Confirmation of the dependencies between the recovery environment, network, AUTH01, DB01, applications and file services.
4. **Backup protection evidence:** Proof of independent access controls, encryption, retention protection and successful restoration from protected copies.
5. **Staffing and operational readiness:** A realistic plan for the two available technicians, including documented responsibilities, procedures and training.
6. **Complete cost information:** Confirmation of recurring and one-off costs, including storage, connectivity, licensing and supplier support.
7. **Business impact evidence:** Historical incident records, credible outage scenarios and validated estimates of the financial and operational effects of downtime.

These checks will help Northbridge determine whether the supplier's quoted performance can be demonstrated in practice and whether the package is suitable for the institute's needs.

Conclusion

Based on the information provided, I recommend the Hot recovery package because it has the shortest estimated recovery time and the smallest stated data-loss window while remaining within the NGN 3,000,000 annual budget. However, it uses the full budget and does not automatically guarantee that every business function will meet its proposed RTO and RPO.

The decision should be conditional on successful recovery testing, evidence from the supplier, adequate staffing, independent protected backups and documented acceptance of any remaining risks. This will allow Northbridge to make a more informed investment decision and avoid relying solely on estimated recovery times.

In the end, the most suitable recovery package is not simply the one with the fastest quoted recovery time. It is the one whose capabilities can be demonstrated, whose risks are understood, and whose cost and recovery performance are acceptable to the institute.

Task 4 — Business Continuity and Disaster Recovery Plan (20 marks)

BUSINESS CONTINUITY PLAN

4.1 Introduction

The flooding and power failure at Northbridge Skills Institute have affected more than the availability of its main office. The incident has disrupted teaching and examinations, fee confirmation, admissions, payroll processing and access to important learner records. Since the main site and its local systems are expected to remain unavailable for 72 hours, the Institute needs to find a way to continue its most important activities while the technical team works on restoring the systems.

My approach would be to manage the incident in two ways at the same time. First, I would put temporary arrangements in place so that essential services can continue as much as possible. Second, I would work with the IT team to recover the affected systems in a controlled manner. Restoring a server alone is not enough; the recovered service must also be secure, contain reliable data and be usable by the staff and learners who depend on it.

The plan covers the five affected business functions, the responsibilities of the teams involved, the recovery steps, communication with stakeholders and the eventual return to the main site. It also recognises that Microsoft 365 email and identity services remain available through mobile internet, while the local applications depend on the separate local directory, AUTH01. This means that having access to email does not automatically mean staff can access the Institute's local applications.

4.2 When the Plan Should Be Activated

The plan should be activated when the incident team confirms that the main site or critical local systems are unavailable and the disruption is likely to affect the Institute's agreed recovery targets. In this scenario, flooding and power failure have already made the main site inaccessible, and the expected disruption is 72 hours. These conditions justify activating the plan without waiting for every affected service to fail separately.

The Incident Manager should coordinate the response and recommend activation to the authorised management decision-maker. If the Incident Manager cannot be reached, the designated deputy should take over so that the response is not delayed. The Institute should formally assign these roles before an incident occurs and make sure that staff know who has the authority to approve the activation.

Once the plan is activated, the Incident Manager should bring together the IT recovery team and the coordinators responsible for the affected business functions. The first discussion should establish what is known, what remains uncertain, which services are most urgent and what can realistically be achieved with the available resources.

For example, the examination scheduled for 11:00 on Day 0 requires immediate attention. The fee confirmation deadline is 16:00, admissions closes at 18:00, and payroll must be submitted to the bank by 12:00 on Day 1. These deadlines mean that the Institute cannot simply wait for all its systems to return before making decisions.

The team should also agree on how often updates will be shared, where decisions will be recorded and how staff should report new problems. Microsoft 365 can support coordination and communication through mobile internet, but it should not be treated as a replacement for AUTH01 or the local applications.

4.3 Roles and Responsibilities

A successful recovery depends on everyone knowing what they are expected to do. The two DR technicians will be responsible for technical restoration, while the business coordinators will help their departments continue working and confirm whether the recovered services meet their needs.

The Incident Manager will coordinate the overall response, manage priorities and escalate issues that require management decisions. The authorised management decision-maker will approve activation and major decisions affecting the Institute, such as significant changes to examination arrangements or the return to the main site.

The IT recovery team will restore the recovery network, AUTH01, DB01 and the affected applications or file services. It will also check that the restored systems are working properly and that access is appropriately controlled. Since only two DR technicians are

available, the recovery schedule must take account of their workload rather than assume that every system can be restored at the same time.

The business coordinators will communicate with learners and staff, organise temporary arrangements and confirm whether business activities can safely resume. They will also help identify missing or inconsistent records after restoration. The finance coordinator, for instance, will need to reconcile payments made through the external payment provider with the records available in FIN01 before confirming outstanding fees.

The following RACI matrix clarifies the main responsibilities. In the matrix, **R** means Responsible for carrying out the work, **A** means Accountable for the outcome, **C** means Consulted, and **I** means Informed. Each activity has one accountable role.

Activity	Accountable	Responsible	Consulted	Informed
Activate the continuity plan	Authorised management decision-maker	Incident Manager	IT recovery lead, business coordinators	Staff and relevant stakeholders
Arrange temporary examinations	Academic/Examination Lead	Examination coordinator	Incident Manager, IT recovery lead	Affected learners and teaching staff
Validate backup availability and integrity	IT Recovery Lead	DR technicians	Security/backup administrator	Incident Manager
Restore the recovery network and AUTH01	IT Recovery Lead	DR technicians	Security/backup administrator	Incident Manager
Restore DB01 and application services	IT Recovery Lead	DR technicians	Relevant business coordinators	Incident Manager
Issue learner and staff updates	Incident Manager	Communications/learner-support coordinator	Business coordinators	Learners and staff
Reconcile finance and	Relevant Business Function Lead	Finance or admissions staff	IT recovery team	Incident Manager

admissions records				
Respond to a suspected security incident	Security/Incident Response Lead	Designated technical responders	IT Recovery Lead, Incident Manager	Authorised management
Approve return to the primary site	Authorised management decision-maker	Incident Manager and IT Recovery Lead	Facilities, security and business leads	Staff and learners
Complete final reconciliation and close the incident	Incident Manager	IT Recovery Lead and business coordinators	Relevant management and service owners	Affected stakeholders

The Institute should confirm the actual job titles and named individuals before approving the plan. Where a role is not currently staffed, management should formally assign responsibility rather than leave an important decision to chance.

4.4 Temporary Arrangements for the Five Business Functions

While the IT team works on recovery, the Institute should continue activities that can be handled safely without the affected applications. Temporary arrangements must be documented, access-controlled and capable of being reconciled with the official records later. They should not create a second, uncontrolled set of records that becomes difficult to merge after restoration.

F1: Teaching and Assessment

The examination scheduled for 11:00 on Day 0 is the most immediate operational concern. The alternate facility has only 25 workstations, while 40 of the 120 learners sitting the examination require Institute-provided workstations. The remaining 80 learners can use their own devices, provided the examination rules, connectivity and security requirements allow it.

The examination coordinator should first confirm whether the learning management system, LMS01, can be made available through the recovery environment in time. The DR technicians should provide an honest update on the likely recovery time rather than promise that the system will be ready before the examination begins.

If the system cannot be restored in time, the Examination Lead should decide whether an approved alternative assessment arrangement is possible. This could involve a controlled rescheduling or another formally approved method, depending on the examination rules.

Learners must receive the same clear instructions, and any alternative must protect the fairness and integrity of the assessment.

The 25 workstations should be allocated according to an agreed arrangement, not on a first-come, first-served basis. The Institute should also confirm that learners using personal devices can access the assessment securely and that their work can be submitted and recorded reliably. If these conditions cannot be met, rescheduling is preferable to conducting an assessment that could disadvantage some learners.

The examination coordinator should keep a record of attendance, approved changes, submissions and any technical issues. These records can then be checked against LMS01 after restoration.

F2: Fees and Finance

The finance function needs to confirm fees by 16:00 on Day 0. The external payment provider remains available, but local receipts and wallet records cannot be updated while FIN01 and DB01 are unavailable. This creates a risk that a learner may have paid successfully but still appear to have an outstanding balance.

For the immediate period, the finance team should use the payment provider's authorised reports or transaction references to maintain a controlled list of payments received. Each entry should include the information needed for later reconciliation, such as the transaction reference, amount, date and relevant learner or account details.

Staff should not promise that every payment has been posted to the Institute's official ledger while the local system is unavailable. Instead, they should explain that the payment is being verified and that the official record will be updated after the system is restored.

Access to the temporary register should be restricted to authorised finance staff. Once FIN01 is available, the team should compare the temporary list with the restored records and the provider's transaction data. Any missing, duplicate or inconsistent entries must be investigated before balances are confirmed. This approach reduces the risk of wrongly marking a learner as unpaid or recording the same payment twice.

F3: Admissions

Admissions closes at 18:00 on Day 0, so the team needs a way to receive and track applications even if REG01 is unavailable. The admissions coordinator should confirm the deadline and any rules governing the receipt of applications, then agree on an approved temporary process with management.

Where permitted, applications and supporting documents can be received through an authorised channel that remains available. The team should issue an acknowledgement to

applicants and maintain a controlled register showing the date received, applicant reference, required documents and current status. Sensitive documents should not be circulated through informal group chats or personal accounts.

The temporary process must not be presented as if the application has already been fully processed in REG01. Staff should explain that the application has been received and will be entered into the official system when it becomes available. If the disruption affects the Institute's ability to meet the published deadline, management should decide whether an extension or other adjustment is necessary and communicate it consistently to all applicants.

After REG01 is restored, the admissions team should enter or reconcile the temporary records, check for duplicate applications and confirm that all submissions received during the outage have been captured.

F4: Payroll

Payroll requires careful handling because the bank submission deadline is 12:00 on Day 1. HR01 and DB01 must be restored, or an approved alternative must be available, before the payroll team can safely complete the process.

The payroll coordinator should establish which stage of payroll processing had been completed before the incident and identify the records needed to prepare the bank submission. If the official payroll system remains unavailable, the team should use only an approved contingency procedure supported by verified source records and the required authorisations.

Payroll instructions should not be created from memory or from unverified spreadsheets. Before any file is submitted to the bank, the team must check employee details, account information, amounts, duplicate entries and approval status. If the Institute cannot verify the information and obtain the required approvals before the deadline, management should contact the appropriate bank representative and explain the situation rather than submit potentially incorrect instructions.

Once HR01 is restored, the payroll team should compare the final submission with the official payroll records and retain evidence of the checks and approvals. Any corrections should follow the Institute's normal authorisation process.

F5: Records and Certificates

The records and certificates function has a 48-hour internal service commitment. Although it may not be as urgent as the examination or payroll deadline, delays can still affect learners who need evidence of their studies or qualifications.

While FS01 is unavailable, the records coordinator should acknowledge requests and maintain a restricted register showing the request date, learner reference, type of document requested and current status. Staff should explain that processing may take longer because the official records system is temporarily unavailable.

The team should not issue a certificate or confirm an academic record solely on the basis of an unverified temporary list. Where an urgent request can be checked against an independently authorised source, the records lead should follow the Institute's approval procedure before responding.

After FS01 is restored, the team should confirm that files are complete, readable and accessible only to authorised users. Pending requests should then be processed in order of urgency, while maintaining the necessary confidentiality and audit trail.

4.5 Disaster Recovery Runbook

The technical recovery process should follow a clear order because the affected systems depend on one another. Restoring an application before its required network, authentication service or database is available will not make the service usable.

The Institute should use the following sequence, with the IT Recovery Lead coordinating the two DR technicians.

Step	Action	Responsible role	Completion check
1	Confirm the incident, record the time and establish which systems are unavailable.	Incident Manager and IT Recovery Lead	Incident record created and affected services identified.
2	Confirm that the alternate recovery facility is safe, powered and suitable for the work.	Facilities lead and IT Recovery Lead	The site and equipment are safe to use.
3	Establish the recovery network.	DR technicians	Required network connectivity is available and tested.
4	Review backup availability, access permissions and any signs of unauthorised changes.	IT Recovery Lead and backup/security administrator	Backup sources are identified and access is appropriately controlled.
5	Restore AUTH01.	DR technicians	Required authentication functions are working and access checks pass.

6	Restore DB01 using a validated backup chain.	DR technicians	The database is accessible, integrity checks pass and the recovery point is recorded.
7	Restore LMS01 and test teaching and assessment functions.	DR technicians and examination coordinator	Authorised users can access the service and complete agreed test activities.
8	Restore FIN01 and test finance functions.	DR technicians and finance coordinator	Application access and essential finance checks pass.
9	Restore REG01 and test admissions functions.	DR technicians and admissions coordinator	Application access and essential admissions checks pass.
10	Restore HR01 and test payroll functions.	DR technicians and payroll coordinator	Required payroll records are accessible and validation checks pass.
11	Restore FS01 and check records and file access.	DR technicians and records coordinator	Required files are readable and access restrictions work as intended.
12	Complete business checks and reconcile temporary records.	Relevant business coordinators	Business owners confirm that essential activities can safely resume.
13	Communicate which services are available and any remaining restrictions.	Incident Manager	Staff and learners receive an accurate update.
14	Continue monitoring and document outstanding issues.	IT Recovery Lead and business coordinators	Issues, decisions and follow-up actions are recorded.

The sequence is a guide, not a reason to ignore urgent business needs. For example, FS01 depends on AUTH01 and the security and backup checks, but it does not depend on DB01 in the scenario. It may therefore be possible to restore it while DB01 is being restored, provided the prerequisites have been met and a DR technician is available.

The recovery schedule must also recognise that only two technicians are available. The team should not start more recovery actions than it can safely manage, and actions should only run in parallel when their dependencies have been satisfied and the required staff are free. The runbook should be updated if actual recovery times differ from the estimates.

Backup Selection and Validation

Before restoring DB01, the team should confirm that the required backup files are available and usable. The scenario identifies B01 as the full DB01 backup from 23:00 on

Day −1 and B02 as the incremental chain up to 07:00 on Day 0. The incremental chain depends on B01 and every required incremental backup being present and valid.

The 07:00 checkpoint is the latest stated recoverable business-data point for DB01, but the team must not assume that the chain is valid simply because the files exist. It should check the sequence, integrity and successful completion of the backups before relying on them.

The AUTH01 snapshot is from 00:00 on Day 0, while the FS01 full backup is from 23:00 on Day −1. These points may leave gaps between the last available copy and the time of disruption. The team should document the potential gap and identify which records may need to be reconciled. The existence of a gap does not automatically mean that every record created during that period has been lost; that must be established through validation.

The disconnected full backup from 17:00 on Day −3 provides an additional recovery option if more recent copies prove unusable. However, using an older backup could mean losing more recent changes, so it should be considered only after assessing the available alternatives and the business impact.

There is also a concern that production and network backup administration share an administrator identity. This makes it important to check whether the backup environment could have been affected by the same compromised credentials or unauthorised changes. The incident team should verify the integrity and access history of the available backups before restoring them.

Testing Before a Service Is Declared Available

Once a system has been restored, the IT team should test more than whether it starts successfully. Technical checks should confirm that the service is reachable, its dependencies are working, relevant logs show no unexplained errors and access controls remain in place.

The relevant business coordinator should then carry out a small number of agreed checks. For LMS01, this could include confirming that an authorised user can access the assessment and that a test submission can be recorded. For FIN01, the finance coordinator should check access to essential records and confirm that the reconciliation process can proceed. Payroll and admissions should undergo similar checks before their teams resume normal processing.

A service should be declared available only when the technical checks and the necessary business checks have passed. The Incident Manager should communicate the service status, any restrictions and what staff are expected to do next.

If a restore fails, the team should stop and investigate rather than repeatedly applying an unverified backup. It should record the error, preserve the available copies and consider

another validated recovery point. Where the failure suggests possible unauthorised activity or data corruption, the security response process should also be considered.

4.6 Communication During the Incident

Clear communication is important because learners and staff need to know what is happening, what they should do and which services they can rely on. The Institute should avoid giving estimated recovery times as firm promises when the technical team has not confirmed them.

Stakeholder	Information to communicate	Channel	Timing	Owner
Learners sitting the examination	Whether the examination will proceed, where to report and what arrangements apply.	Official email, approved learner channels and direct notices where necessary	Immediately, with a further update before the examination	Examination coordinator
All learners	Which services are affected, how to submit urgent requests and when the next update will be issued	Microsoft 365 email and approved communication channels	At activation and at agreed update intervals	Communications/learner-support coordinator
Staff	Available services, temporary procedures, access restrictions and escalation contacts	Official email and staff briefings	At activation and whenever instructions change	Incident Manager
Finance and admissions teams	How to record payments and applications safely and how reconciliation will work	Direct team briefing and approved written instructions	Before temporary procedures begin	Relevant business coordinators

Payroll team and management	Payroll deadline, verified recovery status and any decision required if the deadline is at risk	Restricted official communication	As soon as the risk becomes clear	Payroll coordinator and Incident Manager
Senior management	Current impact, recovery progress, key risks, decisions required and likely next steps	Situation reports and management briefings	At agreed intervals and on material changes	Incident Manager
External payment provider or bank, where necessary	Specific transaction or deadline issues requiring their support	Approved business contact channels	As required	Finance or payroll lead

The Incident Manager should keep a simple situation report containing the time of the update, affected services, work completed, current risks, decisions taken, outstanding actions and the next planned update. This gives management a consistent picture of the incident and reduces the likelihood of different teams sharing conflicting information.

Sample Notice to Learners

Subject: Temporary Disruption to Northbridge Skills Institute Services

Dear Learners,

Northbridge Skills Institute is currently experiencing a disruption following flooding and a power failure at our main site. Some of our learning and administrative systems are temporarily unavailable while the relevant teams work to restore services.

We understand that this may affect examinations, fee confirmation, admissions and requests for academic records. Our immediate priority is to put appropriate arrangements in place for urgent activities and keep you informed as the situation develops.

Learners scheduled for the 11:00 examination should please check the official communication channels for confirmed instructions before proceeding. We will communicate any approved changes to the examination arrangements directly. If you have

made a payment or submitted an application, please retain your transaction reference or acknowledgement while we work to reconcile records after the affected systems become available.

Please use the Institute's approved communication channels for enquiries and avoid sending sensitive personal or payment information through unofficial channels.

We apologise for the inconvenience and appreciate your patience. Further updates will be shared as reliable information becomes available.

Kind regards,
Northbridge Skills Institute Management

4.7 When a Security Incident Must Be Considered

The available information confirms flooding and power failure, but it does not confirm a cyberattack. The Institute should therefore avoid treating the physical disruption as proof that a security incident has occurred. At the same time, the recovery team must not overlook warning signs simply because the incident began with a physical event.

The unexpected backup administrator logon described in the scenario would require investigation, particularly because production and network backup administration share an administrator identity. Other warning signs could include unexplained changes to backup files, unexpected privilege changes, suspicious authentication activity or evidence that restored data has been altered.

If such signs appear, the team should notify the designated Security/Incident Response Lead and preserve relevant logs and evidence. Access to affected accounts or systems may need to be restricted under the approved incident response procedure. The team should avoid making unnecessary changes that could destroy evidence or allow suspicious activity to continue.

The security lead should work with the IT Recovery Lead to determine whether the issue is a genuine security incident, an operational error or something else. Where backup integrity is uncertain, the team should not restore that copy into the recovery environment until it has been assessed. Management should be informed of material risks and decisions that could affect recovery.

This approach allows the Institute to continue dealing with the confirmed flood and power failure while remaining alert to a separate security problem if evidence emerges.

4.8 Returning to the Main Site

The end of the expected 72-hour disruption should not automatically trigger a return to the main site. The building must first be declared safe for use. Facilities staff should confirm that the flood damage has been assessed and that electricity, network connections, environmental conditions and other necessary services are safe and reliable.

The IT team should inspect the affected equipment and infrastructure before reconnecting anything to production. Systems that may have been exposed to water damage or unstable power should not be brought back online simply to speed up the return. The recovery environment should remain available until management is satisfied that the primary site can support the required services.

Before transferring records created during the disruption, the team should take and protect a copy of the current recovery data. The business coordinators should then compare the temporary registers with the restored official records. Finance should reconcile external payment transactions, admissions should check applications received during the outage, payroll should verify submitted instructions and the records team should account for pending requests.

The transfer of new or updated records must be controlled so that the same transaction or application is not entered twice. Where a discrepancy is found, the responsible business owner should investigate it and document the correction rather than silently changing a record. The IT team should also confirm that the systems are synchronised correctly and that users are accessing the intended production environment.

Once the technical checks and business reconciliations are complete, the IT Recovery Lead should recommend whether the Institute is ready to return to normal operations. The authorised management decision-maker should approve the return, based on evidence that the site is safe and essential services are functioning.

Staff and learners should receive a clear notice explaining when normal services resume and whether any restrictions remain. The recovery environment and protected backups should not be decommissioned until the team has confirmed that the primary services are stable and the necessary records have been transferred and checked.

4.9 Conclusion

The Northbridge Skills Institute scenario shows why disaster recovery must cover both technology and day-to-day operations. While the IT team works to restore the network, authentication, database and application services, the business teams must also make practical arrangements for examinations, payments, admissions, payroll and academic records.

The plan therefore combines clear responsibilities, temporary procedures, a dependency-based recovery sequence, regular communication and checks before services are declared available. It also recognises the need to investigate suspicious activity if evidence arises, rather than assuming either that a cyberattack has occurred or that security risks can be ignored.

Most importantly, the Institute should return to its main site only when the environment is safe, the systems have been checked and records created during the disruption have been properly reconciled. This will help the Institute resume normal operations while reducing the risk of further disruption, incorrect records or avoidable harm to learners and staff.

Task 5 — Recovery Schedule and Exercise (15 marks)

5.1 Working out the recovery schedule

For this task, I used the warm-site timings provided in the assignment to determine how quickly Northbridge Skills Institute could bring its five business functions back into operation. I started from 08:00 on Day 0, when the disruption occurred, and followed the dependencies given in the scenario.

One thing I had to consider carefully was the number of technicians available. The Institute has only two DR technicians, and each recovery activity occupies one technician for its full duration. This means that although some activities can run at the same time, they cannot all be started whenever the team wants. The required dependencies must be completed first, and a technician must be available.

The schedule below is my model of how the recovery could run if the stated timings hold and the available backups pass their integrity checks. It is not the result of an actual recovery test.

Recovery action	Technician	Start	Finish	Prerequisite
Restore recovery network	A	08:00	08:30	None
Restore AUTH01	A	08:30	09:00	Recovery network
Complete security and backup checks	A	09:00	09:30	AUTH01
Restore FS01	B	09:30	10:30	AUTH01 and security/backup checks
Restore DB01	A	09:30	11:00	Security and backup checks
Restore FIN01	B	11:00	11:30	DB01

Restore LMS01	A	11:00	11:45	DB01
Restore REG01	B	11:30	12:00	DB01
Restore HR01	A	11:45	12:30	DB01

This arrangement allows the technicians to work in parallel where the dependencies permit it. For example, Technician B can restore FS01 while Technician A works on DB01. Once DB01 is ready, the remaining applications can also be restored without waiting for every other application to finish.

Based on this schedule, all five functions could be technically available by 12:30 on Day 0. However, I would not treat that time as a guarantee. If an integrity check fails, a dependency is unavailable or further validation is needed, the actual recovery time will be longer.

5.2 Can the Institute meet the 11:00 examination?

This is the most important finding from my calculations.

The assessment service depends on five activities being completed in sequence: the recovery network, AUTH01, the security and backup checks, DB01 and finally LMS01.

The total time required is:

30 + 30 + 30 + 90 + 45 = 225 minutes, or 3 hours 45 minutes.

Starting at 08:00, this puts the earliest modelled availability of LMS01 at **11:45 on Day 0**.

The dependency path is:

Recovery network → AUTH01 → Security and backup checks → DB01 → LMS01.

This is the path that controls the assessment recovery time. Even if the team restores another application earlier, that will not make LMS01 available sooner because LMS01 cannot be restored until DB01 is ready.

The problem is that the examination is scheduled to begin at 11:00, while the model suggests that the assessment service will not be ready until 11:45. The Institute would therefore miss the examination start time even if the recovery team follows the schedule successfully.

In my view, this is where business continuity becomes just as important as disaster recovery. The examination coordinator should not wait until 11:00 before deciding what to do. An approved alternative arrangement needs to be prepared in advance, whether that means using another suitable assessment method or rescheduling the examination in line

with the Institute's rules. Any alternative must still be fair to learners and protect the integrity of the assessment.

The calculation also shows why the Institute needs to test its recovery assumptions. A plan may look reasonable on paper, but if the technical recovery time is longer than the time available before an important business deadline, another arrangement is necessary.

5.3 Comparing the recovery times with the RTO and MTD

I compared the modelled recovery times with the RTO and MTD targets proposed in Task 2. The RTO is the target time for restoring an acceptable service, while the MTD is the longest period the business can tolerate the disruption before the consequences become unacceptable.

Function	Modelled recovery time	RTO	MTD	Result
F1 – Teaching and assessment	3 h 45 min	2 hours	3 hours	RTO missed; MTD exceeded by 45 minutes
F2 – Fees and finance	3 h 30 min	4 hours	8 hours	RTO met; within MTD
F3 – Admissions	4 hours	5 hours	10 hours	RTO met; within MTD
F4 – Staff payroll	4 h 30 min	12 hours	28 hours	RTO met; within MTD
F5 – Records and certificates	2 h 30 min	24 hours	48 hours	RTO met; within MTD

The results show that F2, F3, F4 and F5 meet their proposed recovery targets under the modelled conditions. F1 does not. Its recovery time exceeds the two-hour RTO and the three-hour MTD, which reinforces the need for an alternative examination arrangement.

The other functions also need attention, even though their recovery targets are met. Restoring a system within its RTO does not automatically mean that every business deadline has been met.

For instance, FIN01 is expected to be available by 11:30, ahead of the 16:00 fee-confirmation deadline. The finance team would still need to check the restored records and reconcile any payments made during the outage. REG01 is expected to be available by 12:00, giving the admissions team time before the 18:00 closing deadline, but applications received through a temporary process would still need to be checked and entered.

HR01 is expected to be available by 12:30 on Day 0, before the bank submission deadline of 12:00 on Day 1. This gives the payroll team time to validate the information and obtain

the necessary approvals. It should not, however, be taken as permission to submit payroll instructions without checking them.

FS01 is expected to be available by 10:30, but the records team must still confirm that files are intact and accessible to authorised users. Pending certificate and records requests also need to be processed within the 48-hour service commitment.

These results are based on the stated recovery timings. They do not account for unexpected failures, delays or additional work identified during validation.

5.4 What the backup checkpoints tell me

Recovery time is only one part of the problem. The Institute must also consider how much information it could lose when restoring from the available backups. This is where the RPO becomes important.

Using the proposed RPO of 30 minutes for each function, I compared the available checkpoints with the 08:00 disruption time.

Checkpoint	Last available point	Age at disruption	Functions affected	Comparison with 30-minute RPO
B02 – DB01 incremental chain	07:00 Day 0	1 hour	F1, F2, F3 and F4	30 minutes beyond target
B03 – AUTH01 snapshot	00:00 Day 0	8 hours	F1 to F5	7 hours 30 minutes beyond target
B04 – FS01 full backup	23:00 Day −1	9 hours	F5	8 hours 30 minutes beyond target

The DB01 incremental chain provides the latest stated recoverable business-data point, but it depends on B01 and all the required incremental backups being present and valid. For the baseline calculation, I have assumed that the chain passes its integrity checks, as specified in the scenario.

Even with that assumption, the latest DB01 checkpoint is one hour old at the time of disruption. This is 30 minutes outside the proposed RPO. The AUTH01 and FS01 checkpoints are older and leave larger potential gaps.

This does not necessarily mean that all data created during those periods has been lost. It means that the latest available checkpoints do not, by themselves, guarantee recovery to the proposed 30-minute point. The actual missing information would have to be established by checking what changed after each checkpoint and whether reliable records are available elsewhere.

For example, the finance team may be able to reconcile some missing transactions using the external payment provider's records. However, it would be wrong to assume that every missing record can be recreated from those records alone. Any unresolved differences should be investigated and documented.

The results suggest that the Institute needs to review its backup arrangements. More frequent backups may help reduce the gap, but the team must also check that the backup chain is complete, that the copies are protected from unauthorised changes and that restoration has been tested. The proposed 30-minute RPO should not be treated as achieved until the Institute has evidence that its arrangements can support it.

5.5 Why temporary arrangements are still necessary

The schedule shows that four of the five functions could meet their proposed RTOs under the baseline assumptions. However, the Institute still needs continuity arrangements because some deadlines cannot wait for technical recovery.

The examination is the clearest example. LMS01 is expected to be available at 11:45, after the scheduled 11:00 start. The examination coordinator must therefore agree on an approved alternative before the examination begins. The available workstations must also be allocated fairly, especially because 40 learners need centre workstations while only 25 are available.

For finance, the external payment provider remains available, but FIN01 cannot update local receipts and wallet records during the disruption. The team should retain transaction references and other authorised evidence so it can reconcile payments after restoration. This will help prevent learners who have already paid from being wrongly treated as having outstanding fees.

For admissions, a controlled temporary register will help the Institute track applications received while REG01 is unavailable. Once the system is restored, the team should confirm that all applications have been captured and that duplicate records have not been created.

Payroll also needs a contingency arrangement in case HR01 or its validation checks are delayed. The bank submission deadline is 12:00 on Day 1, so there is some time after the modelled restoration at 12:30 on Day 0. However, the team must still verify the information and obtain the required approvals before submission.

Finally, restoring FS01 does not automatically mean that every record or certificate request has been completed. The records team must check the files, confirm access restrictions and work through pending requests within the agreed service commitment.

For me, the main lesson is that the Institute should prepare for the business impact of an outage, not just the technical process of restoring its systems. The two need to work together.

5.6 Tabletop exercise: testing the plan under pressure

I would use a 45-minute tabletop exercise to see how the response team would handle the situation if the original recovery plan no longer worked as expected. The exercise would not involve making changes to live systems. Instead, participants would discuss the decisions they would make, the actions they would take and the information they would need before proceeding.

For this assignment, I would simulate the different roles individually, as permitted. These would include the Incident Manager, IT Recovery Lead, security/incident response representative, examination coordinator, finance coordinator, admissions coordinator, payroll coordinator, records coordinator and authorised management decision-maker.

I would use the following timeline:

Time	What happens
00–10 minutes	Review the initial incident and agree on immediate priorities
10–20 minutes	The main site closure is extended to five days
20–30 minutes	The alternate facility is reduced to 15 working workstations
30–40 minutes	A DB01 incremental backup fails its integrity check, and an unexpected backup administrator logon is reported
40–45 minutes	The payment provider reports successful payments missing from restored finance records, followed by a short review

The exercise would focus on how decisions are made, who is responsible for them and how the changes affect service delivery. The injects below are a separate stress scenario; they do not change the baseline recovery schedule calculated earlier.

Minute 0–10: Establishing the initial response

I would start by confirming the flooding and power failure, the expected 72-hour closure and the systems affected. The Incident Manager would coordinate the response, while the IT Recovery Lead would confirm the technical dependencies and available backups.

Because the modelled LMS01 recovery time is 11:45, the examination coordinator would begin preparing an alternative assessment arrangement immediately. The other business coordinators would also prepare temporary procedures for payments, admissions, payroll and records.

I would make sure the team agrees on how updates will be shared and where decisions will be recorded. This gives everyone a clear starting point and helps avoid confusion when the situation changes.

Minute 10: The site closure extends to five days

My decision: I would continue the alternate-site arrangements and plan for a five-day disruption rather than assume the main site will be available after 72 hours.

Owner: Incident Manager, with approval from the authorised management decision-maker.

What I would do: I would ask facilities staff to confirm whether the alternate site can safely support operations for five days. The business coordinators would review staffing, equipment, learner support and the temporary procedures already in use. The IT Recovery Lead would confirm that the recovery environment and technical support can remain available for the longer period.

Why: The original plan was based on a shorter closure. If the Institute does not adjust its arrangements, it could run short of equipment or support while waiting for the main site to reopen.

Effect: Temporary arrangements and staff workload may need to continue for longer, and management may have to approve additional costs. The team should not return to the main site until it has been cleared as safe and the necessary checks have been completed.

Minute 20: Only 15 workstations are available

My decision: I would ask the Examination Lead to reassess the examination arrangements because the facility can no longer accommodate all 40 learners who need centre workstations.

Owner: Examination Lead, supported by the Incident Manager.

What I would do: The coordinator would identify the affected learners and check whether another suitable venue or approved equipment is available. Learners who can use their own devices should only do so if the examination rules, connectivity, accessibility requirements and security controls allow it. If the Institute cannot provide a fair and secure assessment, management should approve a revised sitting or rescheduling arrangement.

Why: With only 15 workstations available, there is a shortfall of 25. Allocating the available equipment without a clear plan could disadvantage some learners and undermine the integrity of the examination.

Effect: The examination may need to be split into approved sittings or rescheduled. The decision should be communicated consistently to affected learners. This change affects business continuity arrangements; it does not remove the technical dependencies required to restore LMS01.

Minute 30: The DB01 incremental backup fails, and an unexpected administrator logon is reported

My decision: I would not allow the team to use the failed backup chain until its integrity issue had been investigated. I would also escalate the unexpected administrator logon for a security review.

Owners: IT Recovery Lead for the backup decision; Security/Incident Response Lead for the logon investigation.

What I would do: The technical team would preserve the failed backup and relevant logs, check which part of the incremental chain had failed and confirm whether another valid recovery point was available. The disconnected full backup from Day −3 could be considered if more recent options were unusable, but it would involve a greater potential data-loss window.

At the same time, the security team would investigate the administrator logon, including the time, account, source and actions taken. Since production and network backup administration share an administrator identity, I would want to know whether the account could have affected the backups or other recovery resources. Access restrictions would be applied where necessary under the approved incident response procedure.

Why: A backup that fails its integrity check cannot be trusted without further investigation. The unexpected logon does not prove that a cyberattack has occurred, but it raises a concern that should be resolved before the recovery team relies on the affected environment.

Effect: DB01 restoration could be delayed, which would also delay LMS01, FIN01, REG01 and HR01. The team would need to extend temporary business arrangements and update management on the revised recovery estimate. It should not bypass backup validation simply to meet the original schedule.

Minute 40: Successful payments are missing from FIN01

My decision: I would not immediately mark the payments as outstanding or enter them again. I would first reconcile the restored finance records with the payment provider's evidence.

Owner: Finance Lead, with support from the IT Recovery Lead.

What I would do: The finance team would obtain the available transaction references, dates, amounts and confirmation details from the provider. It would compare these with FIN01 and record each missing, duplicate or inconsistent transaction for investigation. Any correction to the official records would be made by authorised staff and supported by an audit trail.

Why: A successful payment may be missing from FIN01 because it was made after the last recoverable database checkpoint. Its absence from the restored system does not automatically mean the payment failed.

Effect: Some fee confirmations may take longer, but the controlled process reduces the risk of wrongly asking learners to pay again or recording a payment twice. Learners whose transactions are still under review should receive a clear explanation.

Minute 45: Closing the exercise

I would close the exercise by summarising the decisions made and the weaknesses identified. The main concerns are the mismatch between assessment recovery and the 11:00 examination, the limited workstation capacity, dependence on a valid DB01 backup chain, shared backup administration credentials and the need to reconcile finance records after restoration.

Each issue should be assigned to an owner with a target date and evidence required for closure. I would not consider the exercise complete simply because the team discussed the problems. The purpose is to turn what we have learned into practical improvements that can be checked later.

5.7 Corrective actions arising from the exercise

The exercise has shown where the Institute needs to strengthen its arrangements. I would give priority to backup integrity and privileged access because problems in these areas could affect several business functions at once.

The dates below are proposed deadlines for this exercise.

No.	Priority	Action	Owner	Target date	Evidence of completion
1	Critical	Test DB01 restoration using a verified backup chain and an alternative recovery point.	IT Recovery Lead	30 October 2026	Signed restore-test report showing the backup used, integrity results and successful database checks.
2	Critical	Separate backup administration from production administration and review the unexpected logon scenario.	Security/Incident Response Lead	30 October 2026	Evidence of separate privileged accounts, approved access controls and a documented logon investigation exercise.
3	High	Update the examination continuity procedure to account for the 11:00 deadline and reduced workstation capacity.	Examination Lead	6 November 2026	Approved procedure covering workstation allocation, alternative arrangements, learner communication and assessment integrity.
4	High	Document and test the reconciliation process for successful payments missing from FIN01.	Finance Lead	6 November 2026	Reconciliation test showing how missing and duplicate transactions are identified, investigated and corrected.
5	High	Review backup frequency and the proposed RPOs for DB01, AUTH01 and FS01.	IT Recovery Lead	13 November 2026	Approved backup schedule, documented RPO review and restore-test evidence for the relevant data sources.
6	High	Confirm that the alternate facility can support a five-day disruption.	Business Continuity Coordinator	13 November 2026	Completed site checklist confirming equipment, connectivity, staffing and learner-support arrangements.

7	Medium	Repeat the recovery scheduling and tabletop exercises after the priority actions have been completed.	Incident Manager	20 November 2026	Exercise report showing recovery timings, decisions, outstanding issues and verified corrective-action closure.

The Incident Manager should maintain a central register of these actions and follow up on progress. An action should only be closed when the agreed evidence has been reviewed. This will help the Institute distinguish between an issue that has genuinely been resolved and one that has only been discussed.

5.8 Conclusion

From the recovery schedule, I found that the Institute could model restoration of all five functions by 12:30 on Day 0, provided the stated timings hold and the backups pass their checks. The main weakness is the assessment service: LMS01 is expected to be ready at 11:45, which is too late for the 11:00 examination and exceeds the proposed F1 MTD.

The tabletop exercise shows how the situation could become more difficult if the closure lasts five days, the alternate facility loses workstations, the DB01 backup fails or payment records do not match after restoration. These are situations where the Institute needs clear decision-making, safe recovery options and practical continuity arrangements.

My conclusion is that recovery should not be measured only by whether a server is running. The service must be secure, the data must be reliable and the people who depend on it must be able to continue their work. The schedule and exercise help identify the gaps in the current plan and give the Institute a practical way to improve before a real disruption occurs.

Task 6 — Governance and Performance Measures

6.1 Ownership and maintenance of the plan

Having a disaster recovery plan is important, but the plan is only useful if people know what to do when a disruption occurs. Northbridge needs clear responsibilities, regular testing and a way to make sure that lessons from incidents and exercises are used to improve the plan.

I recommend that the Business Continuity Coordinator be responsible for maintaining the overall plan, while senior management approves it and provides the resources needed to implement it. The IT Recovery Lead should handle the technical recovery procedures, including restoring servers, checking backups and confirming that systems are working as expected. Heads of the affected business functions should confirm that the restored services meet their needs before normal activities resume. Each key role should also have a named deputy so that recovery does not depend on one person being available.

The plan should be reviewed at least once a year. It should also be updated whenever there is a major change to the institute's systems, suppliers, facilities or business processes, or when an incident or exercise reveals a weakness.

I would use the following testing schedule:

Activity	Frequency	Purpose
Review backup reports and investigate failures	Monthly	Confirm that backups are running and that any failures are addressed.
Restore selected backups in a controlled environment	Quarterly	Confirm that the data can actually be recovered and checked.
Conduct a tabletop exercise	Every six months	Allow staff to discuss their decisions and responsibilities during a simulated disruption.
Carry out a coordinated recovery exercise	Annually	Test the recovery sequence, technical dependencies, staff availability and business sign-off.
Review the plan after a major change or incident	As needed	Ensure that the plan reflects current risks, systems and lessons learned.

Staff involved in recovery should be trained on their responsibilities, communication arrangements and temporary procedures for their departments. New staff taking on critical roles should receive the same training. Exercise results should be documented, with weaknesses assigned to named owners and given target completion dates.

The approved plan should have a version number, approval date, document owner and next review date. Copies should be available to authorised staff, including at a location that remains accessible if the main site is unavailable. Contact details, supplier information and emergency instructions should be checked regularly. The plan and recovery instructions should also be protected so that they remain available during a major IT or security incident.

6.2 Relationship with NIST SP 800-34 and ISO 22301

I used NIST SP 800-34 and ISO 22301 as guides when developing the recommendations in this report. Both support the idea that recovery planning involves more than restoring servers. An organisation also needs to understand its priorities, prepare its people, document recovery procedures, test its arrangements and improve the plan when weaknesses are found.

The table below shows how the work proposed for Northbridge relates to the frameworks discussed in the Week 21 slides. This is a high-level comparison for the assignment and does not mean that Northbridge is certified against either framework.

Northbridge's work	Connection to NIST SP 800-34	Related ISO 22301 theme	Application to Northbridge
Business Impact Analysis (BIA)	Step 2: Conduct a Business Impact Analysis	Clause 4: Context; Clause 6: Planning	Identifies the functions most affected by disruption, their deadlines and the resources they need.
Recovery objectives and backup design	Step 2: BIA; Step 4: Create contingency strategies	Clause 6: Planning; Clause 8: Operation	Helps Northbridge set acceptable recovery times and data-loss limits and select suitable backup arrangements.
Recovery site and contingency arrangements	Step 3: Identify preventive controls; Step 4: Create contingency strategies	Clause 6: Planning; Clause 8: Operation	Provides alternative ways to continue or restore critical services when the main site is unavailable.
Disaster recovery runbook and temporary business procedures	Step 5: Develop the Information System Contingency Plan	Clause 7: Support; Clause 8: Operation	Gives IT staff and business teams clear steps to follow during recovery and while systems are unavailable.
Recovery exercises and staff training	Step 6: Ensure plan testing, training and exercises	Clause 7: Support; Clause 9: Performance evaluation	Checks whether staff understand their roles and whether the recovery arrangements work in practice.
Plan reviews, KPIs and corrective actions	Step 7: Plan maintenance	Clause 9: Performance evaluation; Clause 10: Improvement	Helps Northbridge track performance, address weaknesses and keep its arrangements up to date.

One issue that deserves particular attention is the risk of flooding. Northbridge should check whether the main site, power supply, network equipment and routes to the alternative facility could be affected by the same flood event. Moving recovery activities to another location may not provide enough protection if that location shares the same geographical or infrastructure risks.

The institute should consider geographical separation, alternative connectivity and reliable power at the recovery location. It should also keep a protected backup copy outside the main site and make sure that recovery staff can access it independently. This is particularly important because the scenario states that production and network backup administration share an administrator identity, while the independence of access to the offsite provider has not been documented.

These arrangements should be tested rather than assumed to work. The unexpected backup administrator logon in the exercise should also be investigated before affected systems are trusted. If there is evidence of possible compromise, the security or incident response team should be involved before backups are used to restore production services.

6.3 Key performance indicators and maturity assessment

Northbridge needs a few clear measures to show whether its recovery arrangements are improving. I would focus on whether critical services can be restored within their agreed targets, whether backup restoration tests are completed, whether identified problems are resolved and whether the people responsible for recovery are properly trained.

The following KPIs are proposed for management approval. They are targets for future performance, not current results.

KPI	Formula or measurement	Proposed target	Data source	Reporting frequency	Accountable owner
Recovery performance	Number of critical services restored within their approved RTO ÷ total critical services tested × 100	At least 90%	Recovery exercise and incident reports	After each exercise and quarterly	IT Recovery Lead
Restoration-test coverage	Number of planned restoration tests completed ÷ total restoration tests planned × 100	At least 95%	Backup and restoration-test records	Monthly or quarterly, according to the test schedule	IT Recovery Lead

Corrective-action closure	Number of corrective actions completed by their agreed due dates ÷ total corrective actions due × 100	At least 90%	Corrective-action register	Monthly	Business Continuity Coordinator
Staff training completion	Number of designated staff who completed required training ÷ total designated staff × 100	At least 95%	Training and attendance records	Quarterly, with an annual review	Business Continuity Coordinator
Recovery objectives review	Number of critical functions with current, approved RTO and RPO targets ÷ total critical functions × 100	100%	Approved BIA and recovery plan	Quarterly and after major changes	Business Continuity Coordinator

The results should be reviewed regularly, and any missed target should be investigated. For example, if a backup restoration test fails, the team should record the cause, assign an owner to resolve it and repeat the test to confirm that the problem has been addressed. Similarly, a system should not be counted as successfully recovered simply because its server is running. The data must be checked, access must work correctly and the relevant business owner must confirm that the service is usable.

Maturity assessment

Based on the information provided in the scenario, I would provisionally place Northbridge at **Level 2: Developing (Repeatable)** on the maturity model in slide 13 of the Week 21 teaching deck.

At this level, basic recovery arrangements exist, but testing is not yet regular or consistent enough to demonstrate reliable recovery. Northbridge has some useful arrangements in place, including backups, an alternative facility and two recovery technicians. However, the scenario raises questions about backup independence, administrator access and whether the systems can be restored within the proposed recovery targets.

I would not yet rate Northbridge at Level 3: Defined (Standardized), because there is not enough evidence to show that its recovery processes are consistently documented, regularly tested and supported by clear governance. This is a provisional assessment based on the scenario, rather than a formal audit of the institute.

Over the next 12 months, I would aim to move Northbridge towards **Level 3: Defined (Standardized)**. The focus should be on making the existing arrangements more reliable and demonstrating that they work.

Period	Main actions	Evidence of progress
Months 1–3	Confirm plan ownership, approve recovery targets, review backup access and document recovery procedures.	Approved plan, named deputies, documented access controls and agreed recovery objectives.
Months 4–6	Test the restoration of DB01, AUTH01 and FS01, including data integrity and access checks.	Recorded test results, identified gaps and assigned corrective actions.
Months 7–9	Conduct a tabletop exercise and a coordinated recovery exercise involving IT staff and business users.	Exercise reports showing decisions, recovery times, lessons learned and action owners.
Months 10–12	Review KPI results, close outstanding high-priority actions and update the plan.	Updated plan, KPI report and evidence that agreed improvements have been completed.

By the end of the year, Northbridge should have clearer responsibilities, documented recovery procedures and evidence that its backups and systems can be restored and checked. Management should use the exercise results and KPI reports to decide whether the institute has made enough progress to move to the next maturity level.

The aim is not simply to have a document ready for an audit. It is to make sure that when a disruption happens, staff know what to do, management can make informed decisions and critical services can resume without creating further problems.

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AI-use declaration

I used ChatGPT as a support tool to help organise the assignment, review wording and clarify concepts. I reviewed the material and remain responsible for the analysis, calculations, decisions and final submission. This statement should be retained only if it accurately reflects my use of AI and complies with ICDFA's course policy.